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ORIGINAL ARTICLE

A novel technique based on the improved firefly algorithm coupled with extreme learning machine (ELM-IFF) for predicting the thermal conductivity of soil

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Abstract

Thermal conductivity is a specific thermal property of soil which controls the exchange of thermal energy. If predicted accurately, the thermal conductivity of soil has a significant effect on geothermal applications. Since the thermal conductivity is influenced by multiple variables including soil type and mineralogy, dry density, and water content, its precise prediction becomes a challenging problem. In this study, novel computational approaches including hybridisation of two metaheuristic optimisation algorithms, i.e. firefly algorithm (FF) and improved firefly algorithm (IFF), with conventional machine learning techniques including extreme learning machine (ELM), adaptive neuro-fuzzy interface system (ANFIS) and artificial neural network (ANN), are proposed to predict the thermal conductivity of unsaturated soils. FF and IFF are used to optimise the internal parameters of the ELM, ANFIS and ANN. These six hybrid models are applied to the dataset of 257 soil cases considering six influential variables for predicting the thermal conductivity of unsaturated soils. Several performance parameters are used to verify the predictive performance and generalisation capability of the developed hybrid models. The obtained results from the computational process confirmed that ELM-IFF attained the best predictive performance with a coefficient of determination = 0.9615, variance account for = 96.06%, root mean square error = 0.0428, and mean absolute error = 0.0316 on the testing dataset (validation phase). The results of the models are also visualised and analysed through different approaches using Taylor diagrams, regression error characteristic curves and area under curve scores, rank analysis and a novel method called accuracy matrix. It was found that all the proposed hybrid models have a great ability to be considered as alternatives for empirical relevant models. The developed ELM-IFF model can be employed in the initial stages of any engineering projects for fast determination of thermal conductivity.

Keywords Thermal conductivity · Unsaturated soil · Firefly algorithm · Improved firefly algorithm · Metaheuristic optimisation

Abbreviations

SVR	Support vector regression
GPR	Gaussian process regression
CNN	Convolutional neural network
ANN	Artificial neural network
adj.R^2	Adjusted Coefficient Of Determination
MAE	Mean absolute error
MAPE	Mean absolute percentage error
NS	Nash–Sutcliffe efficiency
PI	Performance Index

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R^2	Coefficient of determination
VAF	Variance account for
WI	Willmott's Index of agreement
WMAPE	Weighted mean absolute percentage error
S	Saturation degree
n	Porosity
γ_d	Dry density
QC	Quartz content
CC	Clay content
SC	Sand content
GSHPs	Ground source heat pumps
k_r	Normalised thermal conductivity
BTES	Borehole thermal energy storage
TDR	Time domain reflectometry
MLAs	Machine learning algorithms
ANFIS	Adaptive neuro-fuzzy interface system
DNN	Deep neural network
OAs	Optimisation algorithms
MOA	Metaheuristic optimisation algorithm
PE	Processing element
FIS	Fuzzy inference system
FF	Firefly algorithm
IFF	Improved firefly algorithm
ELM	Extreme learning machine
MFs	Membership functions
SLFN	Single-layer feed-forward network
m	Population size
itr	Maximum number of iterations
N	Number of neurons
RMSE	Root mean square error
ROC	Receiver operating characteristic
REC	Regression error characteristic
SE	Squared error
AD	Absolute deviation
AUC	Area under the curve
TR	Training
TS	Testing

1 Introduction

Global warming and the increasing consumption of fossil energy resources necessitate the exploration of new resources. Soil is a common geomaterial and can store geothermal energy, particularly in high depths. Renewability, sustainability, and cleanness (i.e. zero emission of CO_2) are the characteristics of the geothermal resource. One can extract geothermal energy through geothermal energy of piles (GEPs) [1, 2]. Furthermore, it provides other applications, including borehole thermal energy storage (BTES), geothermal heated bridge/pavement, ground source heat pumps (GSHPs), and geological carbon dioxide sequestration [3]. The efficiency and performance of such thermo-active ground structures are mainly dependent on the heat storage capacity and thermal conductivity of the geomaterials. In addition, heat transfer is an essential problem of soils in several pressing engineering exercises, e.g. high-voltage cable laying in depths. Thus, the realisation of the heat transfer process of soils and the optimisation of engineering designs require an insight into the thermal properties of the soil. The main thermal properties of soils are thermal capacity, thermal diffusivity, and thermal conductivity (the opposite of thermal resistivity). Thermal conductivity is essential in the analysis of heat transfer and thermal modelling.

The thermal conductivity of soils determines their steady-state heat transfer process. Thus, it plays a crucial role in geothermal applications of soils. The physical properties inherent in the soil determine its thermal conductivity. Furthermore, the volume fraction impacts thermal conductivity [4]. Particularly, the main factors that impact the thermal conductivity of soils are saturation degree (S), porosity (n), dry density (γ_d), soil type [e.g., sand content (SC) and clay content (CC)], soil mineralogy [e.g., quartz content (QC)], packing geometry, and the grains shape.

Several models have been proposed for predicting the thermal conductivity of soils based on the empirical fitness of natural soils to experimental results. For instance, Kersten [5] carried out tests for measuring the thermal conductivity of 19 natural soil types, and empirically related thermal conductivity to the dry density and water content. Johansen [6] introduced the normalised thermal conductivity (k_r) concept to uniquely evaluate the impacts of porosity, soil type, mineral components, and degree of saturation on the thermal conductivity of soils via the relationship between k_r and S. Cote and Konrad [7] investigated thermal conductivity in soils and building materials. They introduced a new relationship between k_r and S that incorporated the variable k to account for the impacts of the soil type. Lu et al. [8] carried out several thermo-time domain reflectometry (TDR) experiments on 12 natural soil types, including silt, sand, loam, and clay loam. They developed a model for thermal conductivity using a new relationship between k_r and S at an extensive range of moisture degrees. Furthermore, Barry-Macaulay et al. [9] proposed a new generalised model for sand-clay mixture thermal conductivity through thermo-TDR probe tests on the basis of k_r.

Despite recent advances, no unified correlation has been developed between thermal conductivity and its influential factors to date. It is also notable that the current mathematical models have been empirically modified to make their application convenient, turning them into semi-empirical (i.e., mixing) models [4]. On the other hand, most empirical and mixing models are developed based on graphical or numerical methods or data regression [10]. Empirical models are superior to mathematical models in terms of process

calculation; however, the results of empirical models are commonly less accurate. The emerging thermal conductivity prediction models of soils suggest that no generalised approach has been developed to provide application simplicity, accurate predictions, and applicability for every soil type. As a result, it is required to develop a novel approach to evaluate the thermal conductivity of soils effectively.

Machine learning algorithms (MLAs) have been used for predicting the thermal conductivity of soils. Rizvi et al. [11] used a deep neural network (DNN) to predict the effective thermal conductivity of sand. Singh et al. [12] developed an artificial neural network (ANN) to estimate the k of moist porous materials. Zhang et al. [13] established an ANN model for predicting k of unsaturated soil. Singh et al. [14] utilised ANN as well as adaptive neuro-fuzzy interface system (ANFIS) for estimating the thermal conductivity of rocks. Zhang et al. [15] assessed the thermal conduction of the soil from cold regions using ANN. Wei et al. [16] used three different MLAs including support vector regression (SVR), Gaussian process regression (GPR) and convolutional neural network (CNN) to predict the k of composite materials. Chen et al. [17] employed GPR for estimating the thermal conductivity of inorganic materials.

To enhance the performance of conventional MLAs, such as ANN and ANFIS, numerous researches have been performed in the field of science and engineering [18–20]. Although traditional MLAs provide better results compared to statistical methods, they are more likely to be caught up in local minima instead of finding the exact global minima, which in turn produces undesirable results. Thus, to alleviate this issue, researchers are now implementing optimisation algorithms (OAs) to update the learning parameters of classical MLAs, which in turn has led to attaining notable results [21, 22]. In this study, a recently developed metaheuristic optimisation algorithm (MOA), called 'Firefly Algorithm' (FF) introduced by Yang [23], is used for predicting the thermal conductivity of soils. FA is a powerful and high-performance nature-inspired OA. Due to its advantages (such as intelligent sub-division and ability of dealing with multimodality), this MOA has widely been used in many science and engineering applications [24, 25]. However, like most of the optimisation algorithms, FA has some limitations such as local minima trapping issue and inability of exploitation in local space, hence, there will be a probability of getting undesirable results in the validation phase. In addition, premature convergence leads to obtaining a local solution instead of a global solution. Therefore, to enhance the performance as well as to explore fireflies, Lévy flight move strategy can be incorporated into FA [40]. The Lévy flight strategy will not only enhance the exploration capability in global space but will perform exploitation in local space. Considering these concepts, this study proposes Improved Firefly Algorithm (IFF) based on Lévy flight move strategy.

In the present study, three well established classical MLAs, including ANN, ANFIS and extreme learning machine (ELM) are coupled with FF algorithm to construct three hybrid models, namely ANN coupled firefly algorithm (ANN coupled firefly algorithm), ANFIS coupled firefly algorithm (ANFIS-FF) and ELM coupled firefly algorithm (ELM-FF), for predicting the thermal conductivity of soils. In addition, the supremacy of IFF is used to construct an improved version of these models, namely ANN coupled Improved firefly algorithm (ANN-IFF), ANFIS coupled Improved firefly algorithm (ANFIS-IFF) and ELM coupled Improved firefly algorithm (ELM-IFF). The ANN, ANFIS and ELM are the most widely used classical MLAs in the field of engineering and sciences, and hence used in the present study to assess the performance as well as generalisation capability of the IFF. For this purpose, six soil parameters including the n, γ_d, QC, S, CC, and SC are used in predicting the thermal conductivity of soils. The experimental results have confirmed the dominance of IFF in all cases in different aspects, such as accuracy, fitness value, and computational cost.

2 Applied machine learning methods

2.1 Classical machine learning algorithms

In this study, three well-known classical machine learning algorithms including artificial neural network (ANN), adaptive neuro-fuzzy interface system (ANFIS) and extreme learning machine (ELM) are used to get hybridised with metaheuristic optimisation algorithms. ANNs are computational systems heavily inspired by the biological neural system and consist of many single units and weighted artificial neurons, forming a neural structure. Through the harnessing of such a structure, ANNs have been employed as effective mathematical instruments for different purposes such as function approximation and pattern recognition. More details of ANN can be found in [26–30]. As a combination of an ANN and a fuzzy inference system (FIS), ANFIS is introduced to surmount the ANN and FIS drawbacks. Fuzzy logic and fuzzy rules are the basis of an ANFIS. They are created in the model's specific training process. Such an inference system involves five layers. More details of ANFIS can be found in [31–33]. ELMs are neural networks and serve as a new innovative data-driven instrument that exploits an advanced single-layer feed-forward network (SLFN) algorithm for producing a solution with a closed-form to the output weights via the least-squares method (once the weights and biases of hidden layer have been fixed). More details of ANFIS can be found in [34–37].

2.2 Firefly algorithm (FF)

The FF was introduced by Yang [23] as a population-based algorithm. A characteristic of fireflies is bioluminescence flashing light, which can be the major courtship signals for the purpose of mating. This is grounded on three idealised flashing characteristic behaviours of fireflies [38] as summarised below.

(a) Every firefly is unisex and attracted by other fireflies, irrespective of their gender.
(b) Firefly has a degree of attractiveness proportional to its brightness (i.e., light intensity). Therefore, for a pair of flashing fireflies, the firefly with lower brightness approaches another firefly with higher brightness. Due to the proportionality of brightness to the distance, higher brightness suggests a lower distance of each pair of fireflies. A pair of flashing fireflies move randomly when they have the same level of brightness.
(c) The objective function that should be optimised determines a firefly's brightness.

Two essential aspects should be considered to properly design the FF (Fig. 1); (1) light intensity I variation and (2) attractiveness β formulation. A firefly's attractiveness is dependent on its light intensity (i.e., brightness), and brightness has a relationship with the objective function. Brightness $I(r)$ monotonically changes with the distance. It is defined as:

$$I(r) = I_0 e^{-\gamma r}, \tag{1}$$

in which I_0 stands for the original brightness, while γ represents the coefficient of light absorption. Due to the proportionality of a firefly's attractiveness to its light intensity to the adjacent fireflies, the firefly's attractiveness β is defined as:

$$\beta = \beta_0 e^{-\gamma r^2}, \tag{2}$$

in which β_0 stands for attractiveness at a distance of zero ($r=0$). The distance of firefly X_i and firefly X_j is shown in the form of a Euclidean distance/:

$$r_{ij} = X_i - X_j = \sqrt{\sum_{k=1}^{k=m} \left(X_{ik} - X_{jk}\right)^2} \tag{3}$$

in which m is the dimension of problem. Firefly i is attracted to firefly j with higher attractiveness. The movements of firefly have three components: (1) the current position of firefly i, (2) attraction to a firefly of higher attractiveness, and (3) a random walk consisting of a randomly generated number (rand − 0.50) in the interval of [0, 1] and the randomisation parameter α. Then, the movement is written as

$$\boldsymbol{X}_{i,t+1} = \boldsymbol{X}_{i,t} + \beta_0 \cdot e^{-\gamma \cdot r_{ij}^2} \cdot \left(\boldsymbol{X}_{j,t} - \boldsymbol{X}_{i,t}\right) + \alpha \cdot (\text{rand} - 0.50). \tag{4}$$

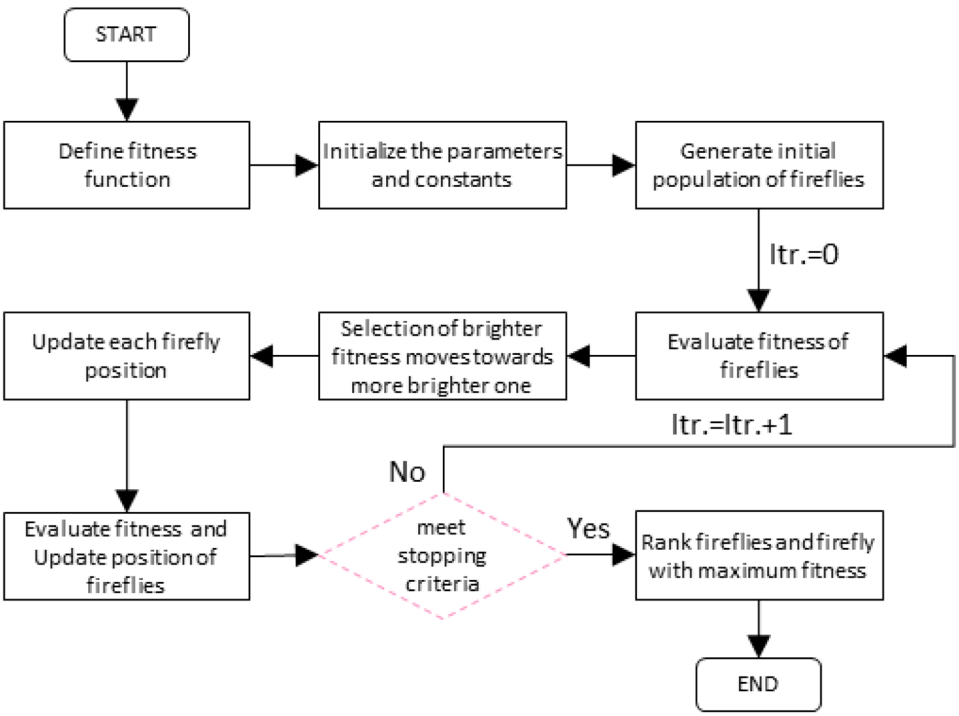


Fig. 1 Flow chart showing the steps of firefly algorithm

2.3 Improved Firefly Algorithm (IFF)

In the case of a high-dimensional optimisation problem, the FF algorithm is often trapped in local optimum [39] which may produce undesirable results. Considering the understanding [40] of 'Lévy-flight-style intermittent scale-free search pattern', Yang [41] incorporates the 'Lévy flight' into FF algorithm to enhance the performance of traditional FF algorithm by boosting the exploration capability in the global space. This process includes randomisation of Lévy distribution instead of traditional uniform distribution and therefore the expression for updating the position $\boldsymbol{X_i}$ of a firefly is given by

$$\boldsymbol{X}_{i,t+1} = \boldsymbol{X}_{i,t} + \beta_0 \cdot e^{-\gamma \cdot r_{ij}^2} \cdot \left(\boldsymbol{X}_{j,t} - \boldsymbol{X}_{i,t}\right) + \alpha \cdot \text{sign}(\text{rand} - 0.50) \cdot \text{Levy}, \tag{5}$$

where α represents the randomisation parameter, and the term sign(rand − 0.50) generates a random direction while the random step length is drawn from the Lévy flights which improve the probability of FF algorithm to jump out of the local optimum and in turn enhances the global search capability of the original FF algorithm. The random number follows the Lévy distribution and can be expressed as;

$$\text{levy}(\eta) \sim \mu = t^{-1-\eta}; (0 \le \eta \le 2), \tag{6}$$

in which, levy(η) is calculated as levy(η) $\sim \frac{\emptyset \cdot \mu}{|v|^{1/\eta}}$, where v and μ follow the standard normal distribution, $\eta = 1.5$, and $\emptyset$ is determined by the following expression

$$\emptyset = \left[\frac{\tau(1+\eta) \times \sin(\pi \times \eta/2)}{\tau((1+\eta)/2) \times \eta \times 2^{(\eta-1)/2}}\right]^{1/\eta} \tag{7}$$

in which the τ represents the standard Gamma function. This Levy-flight-based FF algorithm is proposed as an IFF in this study, and the pseudo code of the IFF algorithm is given in Algorithm 1 showing the incorporation of Lévy flight.

Algorithm 1 ***Pseudo code*** **of FF and IFF algorithm**

Set fitness function $f(x)$ @x, $x = (x_1, x_2, x_3, \ldots, x_D)^T$;
Set Max_gen; # Maximum generation
Initialise $pop_size\ Xi$, dim, ub, lb; $i = 1,2,3, \ldots, n$; # n fireflies
Determine Light Intensity I_i at x_i;
Define γ; # $Light\ absorption\ coefficient$
while ($t < Max_gen$)
    for $i = 1{:}n$;
        for $j = 1{:}n$;
            if $I_i < I_j$ then
                move firefly I to j

            end
            vary attractiveness with r using $exp(-\gamma r)$
            evaluate new solution using Eq. (23) # $for\ Classical\ Firefly\ Algorithm$
            evaluate new solution using Eq. (24) #$for\ Improved\ Firefly\ Algorithm$
            update I_i
            update fitness
        end
    end
    Update Best Solution Ever Found
    Rank the fireflies and find the current global best
end
return best solution
incorporation of Levy Flight
set $\eta = 1.5$;
calculate sigma ($\emptyset$) using Eq. (26);
set u and v value;
calculate $steps$ of levy flight

2.4 Hybridisation process

In the present study, a recently proposed metaheuristic optimisation algorithm, Firefly algorithm, is used to optimise the learning parameters of ANN, ANFIS and ELM for predicting the thermal conductivity of soils. In ANN, the learning parameters are the input weights, biases of hidden neurons, output weights and output bias. In ANFIS, the learning parameters include the antecedent and consequent parameters, and in ELM, the parameters are input weights, hidden biases and output weights. After initialisation of ANN, ANFIS and ELM, the FF algorithm is used to optimise the learning parameters, i.e., the weights and biases of the models. In this regard, the FF algorithm is set up before optimising the learning parameters of MLA, including the population size (m), maximum number of iterations (itr), lower bound, upper bound and other parameters besides number of neurons (N) in the hidden layer of ANN and ELM, and number of fis parameter in ANFIS. Then, the weight and biases of ANN and ELM, and antecedent and consequent parameters of ANFIS are optimised by FF algorithm. A similar procedure as those used in ANN-FF, ANFIS-FF and ELM-FF modelling, is also applied for the ANN-IFF, ANFIS-IFF and ELM-IFF models. The optimised values of learning parameters are determined with the lowest root mean square error (RMSE). Although the number of hidden neurons, population size, maximum number of iterations, lower bound, upper bound and other parameters are kept the same in FF and IFF based optimisation, the optimised value of learning parameters are different. In addition, note that the above mentioned parameters ($n, N,$ itr, lb, ub, etc.) play an important role in the optimisation process, hence, they need to be tuned properly during the course of hybridisation process. The entire process of hybridisation is shown in Fig. 2, representing the steps in developing the ANN, ANFIS and ELM model optimised with FF and IFF algorithm.

3 Data processing and analysis

3.1 Data preparation

In computational analysis, data normalisation is regarded as a crucial step. In the pre-processing stage, data normalisation must be performed to nullify the dimensional effects of the variables. So, the input and output variables are normalised between 0 and 1 before model development employing the formula $\{(x-x_{\min})/(x_{\max}-x_{\min})\}$ known as the 'min–max' normalisation approach. Here, $x_{\min}$ and $x_{\max}$ denote the minimum and maximum values of the parameter (x), respectively. The entire dataset is divided into training (TR) and testing (TS) subsets after the normalisation step. For this reason, 75 percent (193 cases) of the entire dataset is randomly distinguished as the training set, and 25 percent of remnants (64 cases) are considered as the testing set.

In this study, a total of 257 measurements of soil thermal conductivity have been obtained. The compiled database contains the 160, 80, 2, 3, 10, and 2 soil samples used by Tarnawski et al. [42], Chen [43] and Zhang et al. [44], McCombie et al. [45], Tarnawski et al. [46], Tokoro et al. [47], and Tarnawski et al. [48], respectively. To assess the soil thermal conductivity, the reports employed the heat pulse approach (known as the transient state technique). During the experiments, this approach can neglect as much

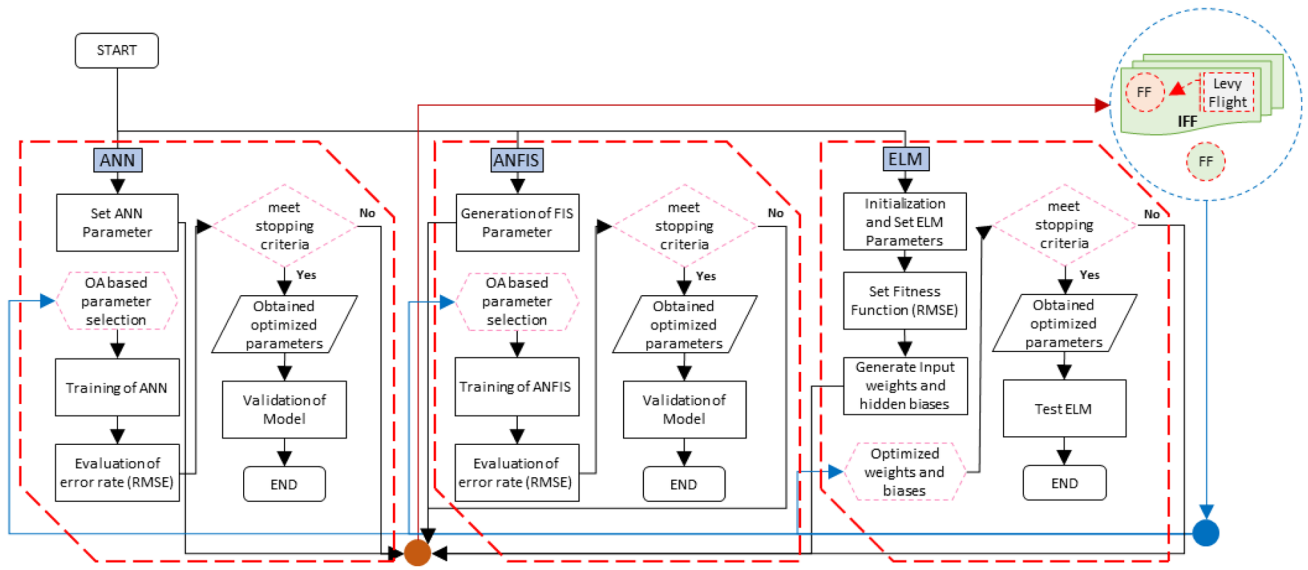


Fig. 2 Flow chart showing the steps in hybrid modelling

Table 1 Statistical details of soil parameters (all soil together)

	γ_d (g/cm^3)	n (–)	S (–)	QC (–)	SC (–)	CC (–)	k (W m^{-1} K^{-1})
Mean	1.4387	0.4637	0.3321	0.6125	0.6199	0.0857	1.1667
StDev	0.1837	0.0665	0.2173	0.3052	0.3855	0.1133	0.6276
Min	0.9800	0.3500	0.0400	0.0000	0.0000	0.0000	0.1000
25%	1.3000	0.4100	0.1000	0.3700	0.2600	0.0000	0.6600
50%	1.4800	0.4500	0.2500	0.6000	0.7700	0.0300	1.1400
75%	1.5900	0.5200	0.5000	0.9900	1.0000	0.1200	1.5700
Max	1.8300	0.6300	0.7000	1.0000	1.0000	0.4200	2.9600

Table 2 Performance measures and their ideal values

Trend indices	Ideal value	Error indices	Ideal value
Adj.R^2	1	R^2	1
MAE	0	RMSE	0
MAPE	0	VAF	100
NS	1	WI	1
PI	2	WMAPE	0

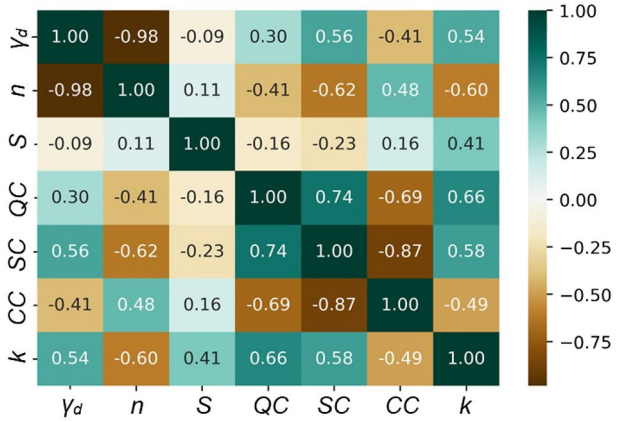


Fig. 3 Correlation matrix of the input and output variables

moisture migration as possible. Thus, the precision of the calculated soil thermal conductivity is assured.

For predicting the thermal conductivity, a collection of 257 datasets including several soil types together with six soil parameters, i.e. γ_d, n, S, QC, SC, and CC, were considered. The soil parameters considered in the study have been presented in Tables 1 and 2.

3.2 Statistical analysis and visualisation

According to Fig. 3, the thermal conductivity is correlated with soil characteristics (input variables) including dry density, porosity, saturation degree, quartz content, sand content, and clay content. The figure indicates the degree that k correlates with the input parameters. It is observed that QC has the maximum correlation (0.66) with k. This is followed by the values of n, SC, γ_d, CC, and S equal to − 0.60, 0.58, 0.54, − 0.49, and 0.41, respectively.

The linear behaviours among k and input parameters are illustrated in scatter plots in Fig. 4a to visualise the influence of the soil parameters on k value. In addition, to better demonstrate the present database, histogram plots are furnished in Fig. 4b. Figure 4a represent that there is no considerable correlation between the six input variables and the output variable, implying that traditional learners like linear regression cannot imitate the nonlinear relations among inputs and outputs. On the other hand, from Fig. 4b, it is indicated that, the present database, the soils mostly have the SC and CC in the range of 0.6–1.0 and 0–0.20, respectively. While the QC lies mostly in the range of 0.1–0.4, 0.5–0.7, and 0.9–1.0. The majority of soils have the saturation degree from 0.05 to 1.5, 0.25 to 0.30, 0.50 to 0.55, and 0.65 to 0.70. The n and γ_d are in the range of 0.3–0.55 and 1.1–1.7 g/cm^3, respectively. And, for the k, it locates in the range between 0.1 and 3.0.

3.3 Description of the performance parameters

Six metaheuristic hybrid models, including ANN-FF, ANN-IFF, ANFIS-FF, ANFIS-IFF, ELM-FF, and ELM-IFF have been developed to calculate the thermal conductivity in unsaturated soils. According to Sect. 3.1, 75% of the entire dataset (193 samples) was used to train, and the remaining part (64 cases) was employed to validate (test) the models.

Ten performance parameters, including Adjusted Coefficient of Determination (adj.R^2), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Nash–Sutcliffe efficiency (NS), Performance Index (PI), Coefficient of Determination (R^2), RMSE, Variance Account For (VAF), Willmott's Index of agreement (WI) and Weighted Mean Absolute Percentage Error (WMAPE), for determining the efficiency were calculated. These performance parameters are defined according to

$$\text{Adj.}R^2 = 1 - \frac{(m-1)}{(m-p-1)}\left(1-R^2\right), \tag{8}$$

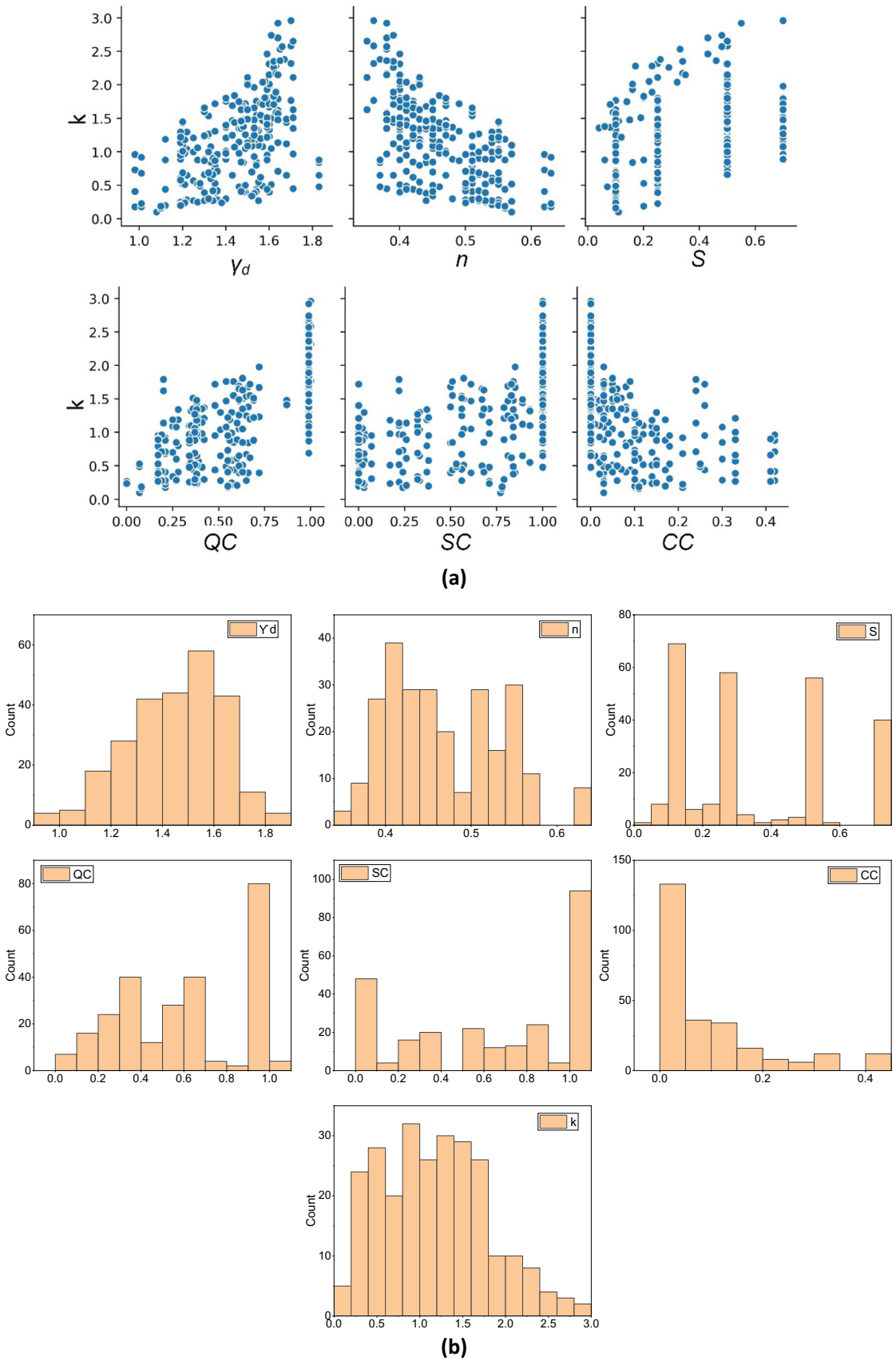


Fig. 4 Illustration of **a** scatter plots and **b** histograms of input and output variables

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{m}\left|\left(\hat{y}_i - y_i\right)\right|, \tag{9}$$

$$\mathrm{MAPE} = \frac{1}{n}\sum_{i=1}^{m}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100, \tag{10}$$

$$\mathrm{NS} = 1 - \frac{\sum_{i=1}^{m}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{m}(y_i - y_{\mathrm{mean}})^2}, \tag{11}$$

$$\mathrm{PI} = \mathrm{adj}.R^2 + 0.01\mathrm{VAF} - \mathrm{RMSE}, \tag{12}$$

$$R^2 = \frac{\sum_{i=1}^{m}(y_i - y_{\mathrm{mean}})^2 - \sum_{i=1}^{m}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - y_{\mathrm{mean}})^2}, \tag{13}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2}, \tag{14}$$

$$\mathrm{VAF}(\%) = \left(1 - \frac{\mathrm{var}(y_i - \hat{y}_i)}{\mathrm{var}(y_i)}\right) \times 100, \tag{15}$$

$$\mathrm{WI} = 1 - \left[\frac{\sum_{i=1}^{m}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}\left\{\left|\hat{y}_i - y_{\mathrm{mean}}\right| + \left|y_i - y_{\mathrm{mean}}\right|\right\}^2}\right], \tag{16}$$

$$\mathrm{WMAPE} = \frac{\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times y_i}{\sum_{i=1}^{m} y_i}, \tag{17}$$

where y_i and $\hat{y}_i$ are the experimental and expected ith values of thermal conductivity, m is the data sample number in a dataset (regardless of training and testing data), y_{mean} represents the average of the experimental values. The value of the parameters should be equal to their ideal value for a perfect predictive model (Table 3). Additionally, the values of the indices are fitted in a way that can enable us to verify the proper ranges for the performance parameters. A thorough understanding of the performance parameters has been presented in the following.

The performance parameters can be classified as parameters for measuring trends (i.e. R^2, adj. R^2, VAF, PI, WI, and NSE), and parameters for measuring errors (RMSE, MAE, RSR, and WMAPE). The parameters in the first group indicate the predictive model trend while the error measuring class quantifies the predictive mode error [49–52]. In statistics, R^2 measures the linear correlation between the actual and expected values, VAF determines the relative variance, PI measures the combined effect of (adj.R^2, VAF, and RMSE), WI explains the degree of model prediction error, and NS measures the relative magnitude of the residual to the actual variance of the data. In contrast, RMSE and MAE are the common indices in a set of predictions which evaluate the precision. RMSE expresses the root mean square differences between the actual and the estimated value, MAE measures the average error magnitude, MAPE calculates the average error percentage and WMAPE estimates the sum of error weighted by the actual output values. Employing these parameters, the degree of correction and the model error can be approximated. However, the higher the value of R^2, VAF, PI, WI, and NS, the stronger the predictive capacity of the model, and the lower the value of RMSE, MAE, MAPE and WMAPE, the better the model is. In the current research, the parameters are evaluated in detail to ascertain the best predictive model and the outcomes are presented in the discussion section.

Table 3 Details of the optimum parametric configuration of the MOAs

Parameters	ANN	ANFIS	ELM
Hidden neurons	10	–	10
fis(ANFIS)	–	10	–
Population size (n)	20	20	20
Iterations (itr)	100	100	100
Gamma	2	2	2
Beta	2	2	2
Alpha	0.2	0.2	0.2
Alpha_damp	0.98	0.98	0.98
lb, ub	− 1, + 1	− 1, + 1	− 1, + 1
cost function	RMSE	RMSE	RMSE

4 Results and discussions

In the current research, the parameters were evaluated in detail to ascertain the best predictive model and the outcomes have been presented in the discussion section. The entire dataset is then categorised into subsets of the train (193 observations) and the test (64 observations). Then the training subset was used to construct metaheuristic models based on ANN, ANFIS, and ELM, while the testing dataset was applied to validate the model. The performance indices were measured and evaluated to approximate the performance of the model predictive. The predictive performances of the developed models are discussed in the following along with a comparative evaluation of the metaheuristic algorithm parametric configuration employed in ANN, ANFIS, and ELM-based modelling; and the best predictive model is determined.

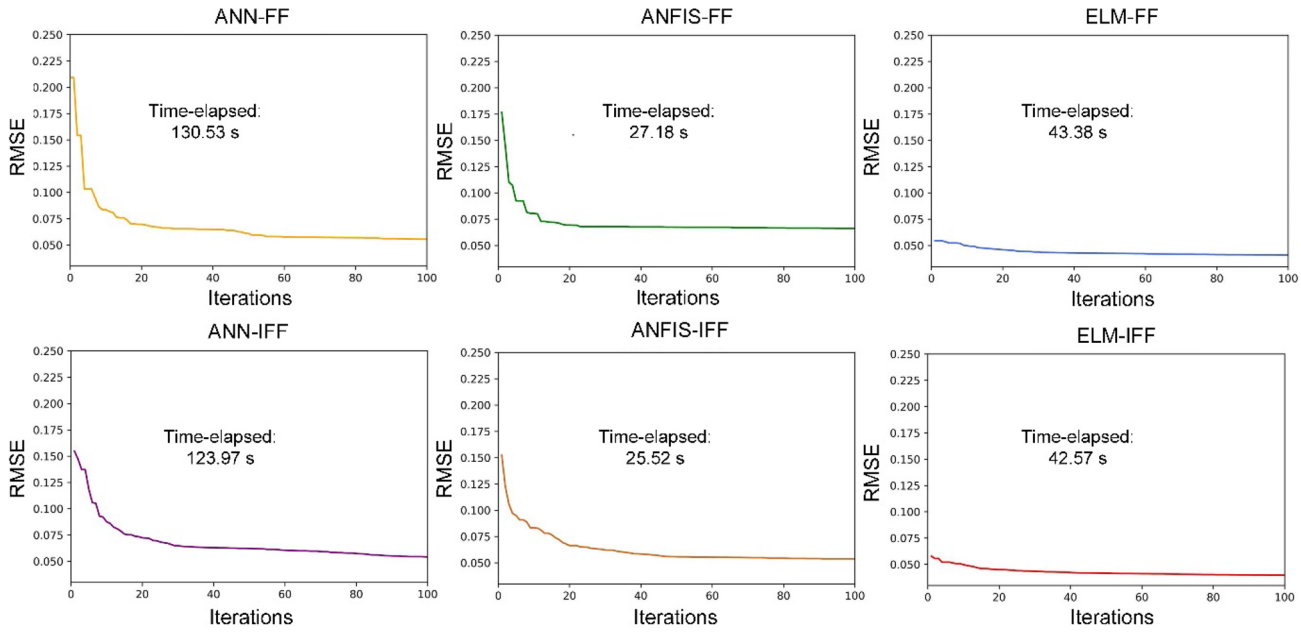


Fig. 5 Iterative performance of the hybrid models

4.1 Parametric configuration of developed models

Two optimisation algorithms, after arranging the experimental data, including FF and IFF, are employed to optimise weights and biases of ANN, ANFIS, and ELM-based models. Because of the randomness of metaheuristic algorithms, identifying the parametric configuration, i.e. adjustable parameters, is crucial in every optimisation modelling. Therefore, the adjustable parameters have to be tuned during the execution of the process. Using the trial-and-error approach, in ANN modelling, the optimum number of hidden neurons in the hidden layer is obtained as 10. In this analysis, robust analyses on deterministic parameters are employed (see Table 3). It is found that the IFF algorithm worked effectively (in the testing step based on the RMSE and R^2 values) in ANN, ANFIS and ELM. The iterative performance is shown in Fig. 5.

4.2 Simulation of different developed models

Metaheuristic optimised ANN, ANFIS and ELM-based models, including ANN-FF, ANN-IFF, ANFIS-FF, ANFIS-IFF, ELM-FF, and ELM-IFF, are developed for predicting the thermal conductivity of soil. After data portioning, the training dataset is employed to train the ANN, ANFIS and ELM utilising FF and IFF algorithms, then the optimal weights and biases are retrieved. One hidden layer with 10 hidden neurons should be considered for the ANN models. The numbers of hidden neurons are chosen based on trial-and-error driven approaches due to their randomness structure. Additionally, the linear (trainlm) and tangent sigmoid (tansig) functions are respectively employed as training and transfer functions. ANN-FF, ANN-IFF, ANFIS-FF, ANFIS-IFF, ELM-FF, and ELM-IFF models are designed based on the details provided in Table 3 The testing dataset is subsequently applied to evaluate the generalisation potential of the models.

The performance index values for models are calculated separately for the training and testing datasets (Tables 4 and 5). In the training stage, the ELM-IFF model achieved the best accurate predictions with $R^2 = 0.9676$, VAF = 96.76, RMSE = 0.04 and MAE = 0.0294, and in the testing phase the values were recorded as $R^2 = 0.9615$, VAF = 0.9606, RMSE = 0.0428 and MAE = 0.0316. After thoroughly comparing the models, the values of other performance indices are illustrated in Tables 4 and 5. The scatter plots for the predictive models (ANN-FF, ANN-IFF, ANFIS-FF, ANFIS-IFF, ELM-FF, and ELM-IFF), in the training and testing phases, are depicted in Figs. 6 and 7, respectively.

4.3 Statistical details of results

This section describes the outcomes of all the performance parameters of the models (i.e. adj.R^2, MAE, MAPE, NS, PI, R^2, RMSE, VAF, WI, and WMAPE). The output parameter values for all six models for the training and testing datasets are reported in Tables 4 and 5, respectively. All models have captured the correlations in predicting the soil compression index. The ELM-IFF model achieved the highest prediction performance in modelling with $R^2 = 0.9676$, VAF = 96.76,

Table 4 Values of performance parameters of the hybrid models for the training stage

Performance parameters	Training set					
	ANN-FF	ANN-IFF	ANFIS-FF	ANFIS-IFF	ELM-FF	ELM-IFF
AdjR^2	0.9355	0.9386	0.9093	0.9394	0.9647	0.9666
MAE	0.0446	0.0411	0.0510	0.0419	0.0303	0.0294
MAPE	17.3861	17.5630	32.0215	21.7236	12.2956	11.6981
NS	0.9376	0.9406	0.9110	0.9414	0.9659	0.9677
PI	1.8174	1.8249	1.7549	1.8268	1.8895	1.8942
R^2	0.9375	0.9405	0.9122	0.9413	0.9658	0.9676
RMSE	0.0555	0.0542	0.0663	0.0538	0.0410	0.0400
VAF	93.7458	94.0461	91.1866	94.1251	96.5817	96.7603
WI	0.9837	0.9845	0.9767	0.9847	0.9913	0.9917
WMAPE	0.1175	0.1083	0.1343	0.1104	0.0798	0.0775

Table 5 Values of performance parameters of the hybrid models for the testing stage

Performance parameters	Testing set					
	ANN-FF	ANN-IFF	ANFIS-FF	ANFIS-IFF	ELM-FF	ELM-IFF
AdjR^2	0.9178	0.9275	0.9070	0.9192	0.9493	0.9575
MAE	0.0479	0.0456	0.0552	0.0457	0.0382	0.0316
MAPE	21.3346	22.6109	25.6125	20.8393	16.6827	13.3534
NS	0.9191	0.9295	0.8989	0.9123	0.9495	0.9579
PI	1.7817	1.8029	1.7406	1.7705	1.8537	1.8753
R^2	0.9257	0.9344	0.9159	0.9269	0.9541	0.9615
RMSE	0.0593	0.0554	0.0663	0.0618	0.0469	0.0428
VAF	92.3253	93.0874	89.9928	91.3118	95.1317	96.0600
WI	0.9797	0.9826	0.9763	0.9794	0.9876	0.9895
WMAPE	0.1347	0.1276	0.1510	0.1203	0.1070	0.0890

RMSE = 0.04 and MAE = 0.0294 in the training phase; the values for the testing phase are recorded as $R^2 = 0.9615$, VAF = 0.9606, RMSE = 0.0428 and MAE = 0.0316. Additionally, PI, WI, NS, MAE, MAPE, and WMAPE values demonstrate no uncertainty in the results because all parameters obey the intended trends.

4.4 Visual interpretation of results

Visualisation of model outcomes is rated as one of the most crucial steps in soft computing studies. Description of raw results in graphical forms usually facilitates spectators to discover the findings to reveal genuine insight. Considering this concept, visual interpretations of findings (Taylor diagram and Regression Error Characteristics graph) are outlined in the following sections with a novel method of accuracy matrices.

4.4.1 Taylor diagram

Taylor diagram [53] is a two-dimensional mathematical illustration targeted at highlighting the precision of different models most suitably and compactly. It is a succinct description of a statistical study that proves how well the findings fit the actual and developed methods. This diagram seems to be a viable technique that can be applied to measure the degree of (dis)similarity between the modelled outcomes and the observations. In this graph, a single point indicates the coefficient of association, the root-mean-square-difference, and the ratio of the standard deviations of the two variables. The point closer to the 'reference point' signifies the perfect predictive model. Figures 8 and 9 display the Taylor graphs for the training and testing sets, respectively, showing the positions of all the models by points (different markers). The ELM-IFF model's position seems to be closer to the 'reference point' and can therefore be regarded as the best one.

4.4.2 Regression error characteristic curve

As a modified regression variant of the receiver operating characteristic (ROC) curve [54], the error tolerance and the precision of regression are plotted on the *x*-axis and *y*-axis in the regression error characteristic (REC) curve [55], respectively. Commonly, the error on x-axis is described in terms

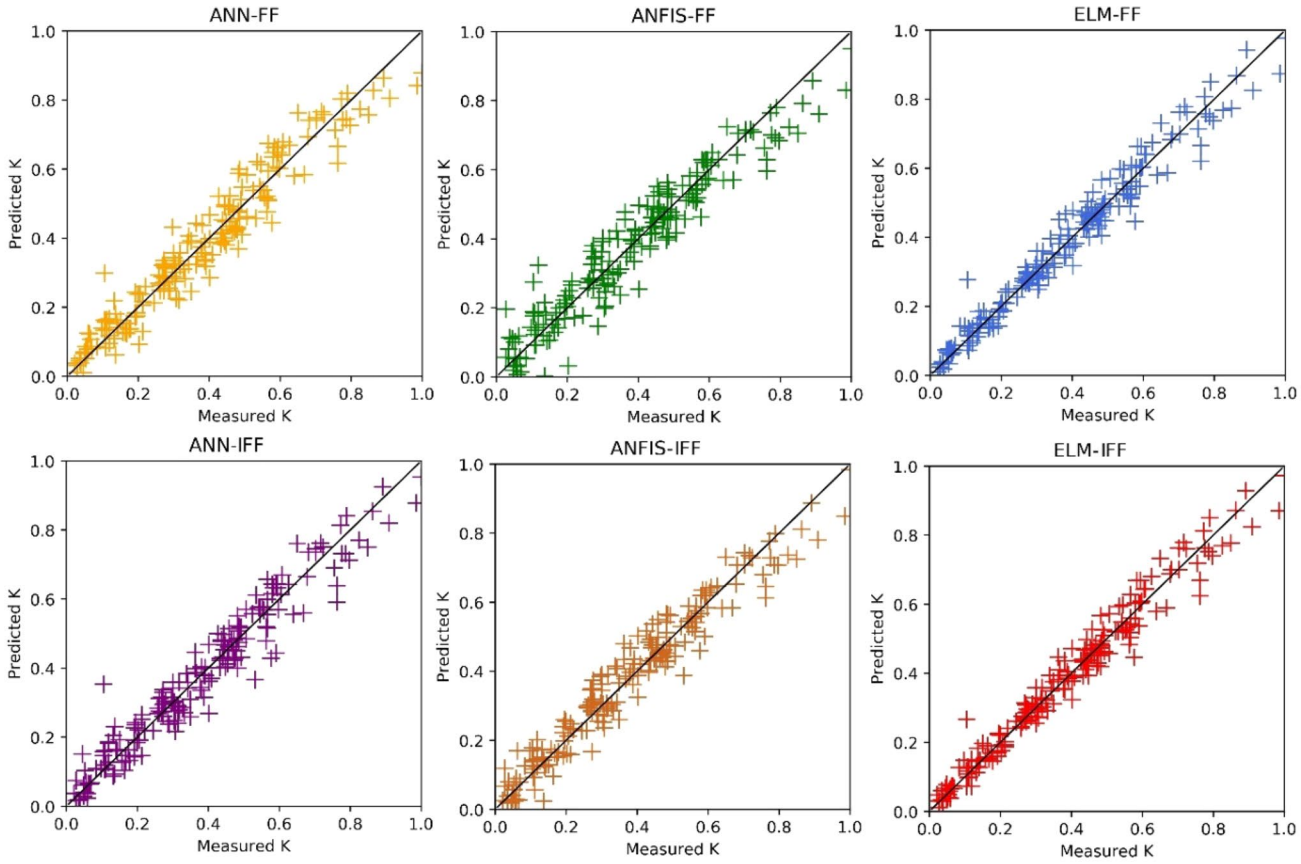


Fig. 6 Regression plots between measured and predicted values for the training stage

of squared error (SE) or absolute deviation (AD). The resulting curve, however, measures the 'cumulative distribution function' of error such as cdf. The closer the corresponding curve position is to the upper left corner, the more accurate is the model. The value of the area under the curve (AUC) is determined to measure the overall precision of the model, and it should ideally be as large as possible to have a proper prediction. This idea of error tolerance is extremely tempting since the probability of inaccuracy is high in most regression data as a result of uncertainties and errors in experiments. In Figs. 10 and 11, the REC graphs for train and test phases are displayed, respectively, for the corresponding models. Additionally, the AUC values have been illustrated in the figures. The AUC values of all predictive metaheuristic models seem to be on the higher side, and the ELM-IFF model, during the train and test stages, reached the maximum AUC value of 0.9443 and 0.9308, respectively; thus, ELM-IFF is the best model.

4.4.3 Accuracy matrix

An accuracy matrix, which is a novel graphical assessment in the form of the heat map, is examined to evaluate the efficiency of the models. The matrix consisted of different statistical parameters to determine the predictive performance of the model for training (TR) and testing (TS) datasets. Figure 12 illustrates the accuracy matrix of six models in terms of predicting the soil thermal conductivity. Accuracy matrix illustrates the obtained accuracies (in %) of the performance parameters by comparing them to their ideal values, according to

$$A_{\mathrm{e}} = \left|\left(1 - |p_{\mathrm{e}}|\right)\right| \times 100, \tag{18}$$

$$A_{\mathrm{t}} = \frac{|p_{\mathrm{a}}|}{i_{\mathrm{a}}} \times 100, \tag{19}$$

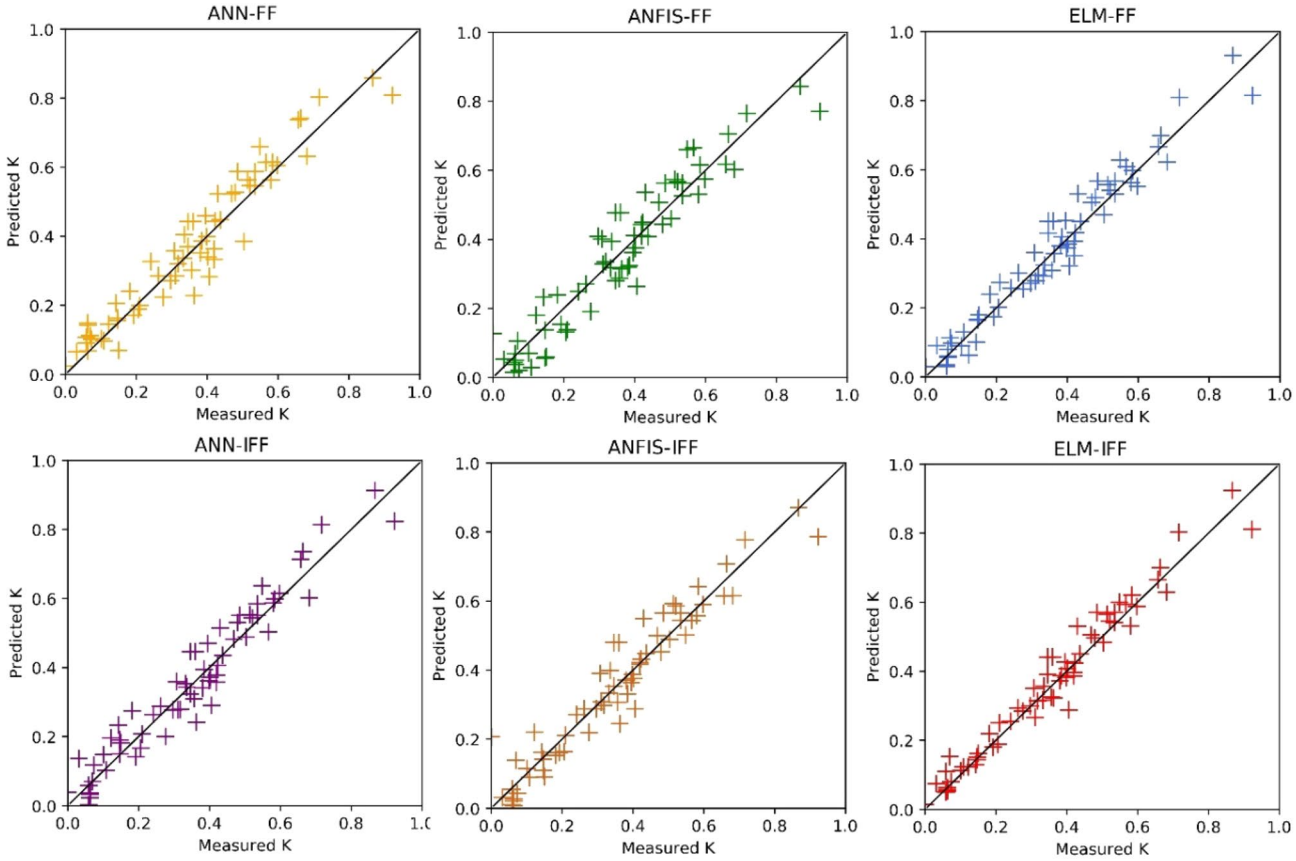


Fig. 7 Regression plots between measured and predicted values for the testing stage

where, A_e and A_t depict the accuracy for the error and trend measuring parameters, i_a is the ideal value of parameters, p_e and p_a show the obtained values of error and trend measuring parameters, respectively. Performance parameters which measure the error (i.e. MAE, MAPE, RMSE and WMAPE) and trend (i.e. adj.R^2, NS, PI, R^2, VAF and WI) are evaluated using Eqs. 18 and 19, respectively.

4.4.4 Rank analysis

To evaluate the performance of the best predictive model, several indices were calculated and conclusions were drawn by comparing the results. This is an intelligent approach, contributing to erroneous outcomes. Thus, "Rank Analysis" can be conducted to corroborate the claims based on index parameters [56, 57]. This is a very efficient and well-organised strategy to easily evaluate the overall performance of a model. According to the analysis, for both training and testing datasets, a maximum rank of m (the total number of corresponding models) was assigned to the ones with the best index values, and a minimum rank equal to one was assigned

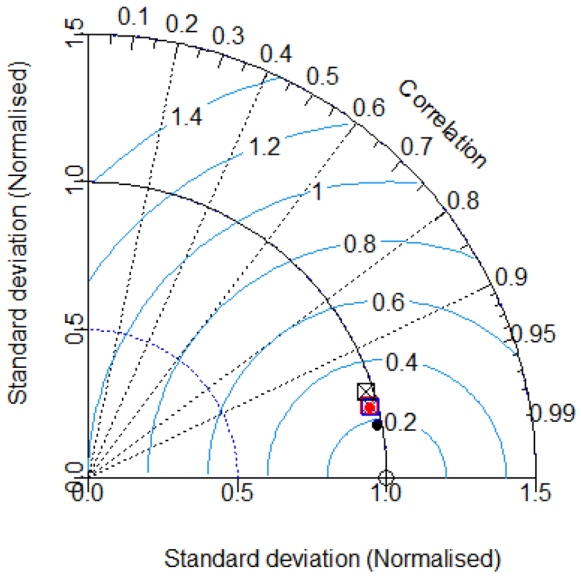


Fig. 8 Taylor diagram for the training dataset

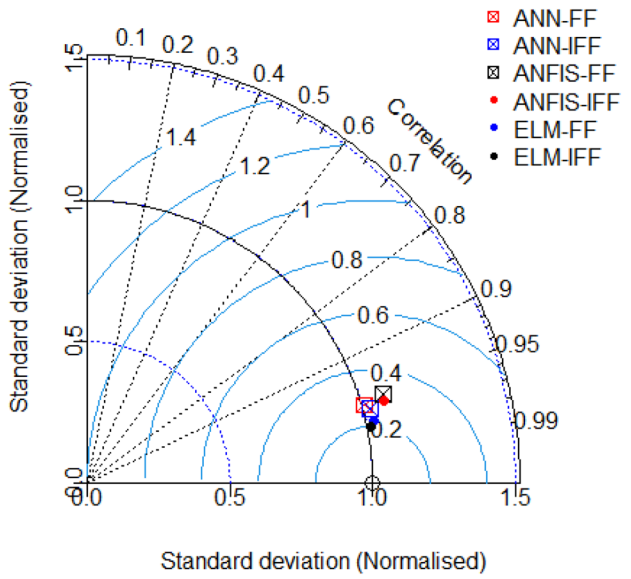


Fig. 9 Taylor diagram for the testing dataset

to the models having the least favourite values. The overall performance (total rank) of all models was determined in subsequent stages by summarising the ranks for every dataset. Eventually, the final rank for every model was estimated by summarising the total ranks of the datasets. Tables 6 and 7 contain the rank analysis for the developed model. The ELM-IFF model, obtained the highest final rank of 120, followed by ELM-FF (rank = 100), ANN-IFF (rank = 69), ANFIS-IFF (rank = 63), ANN-FF (rank = 48). In addition, ANFIS-FF reached the lowest final rank of 20. Throughout both the training and testing phases, the ELM-IFF model reached the highest total rank of 60; while the ANFIS-FF obtained the lowest total rank of 10.

4.5 Sensitivity analysis (SA) and AIC evaluation

SA is a technique that is used to determine if the dependent variable(s) of the model is affected by changes in the independent variables of a numerical model. To perform SA, the widely used Cosine Amplitude Method (CAM) was used in the present study. In CAM, all of the data pairs are expressed in common X-space and used to construct a data

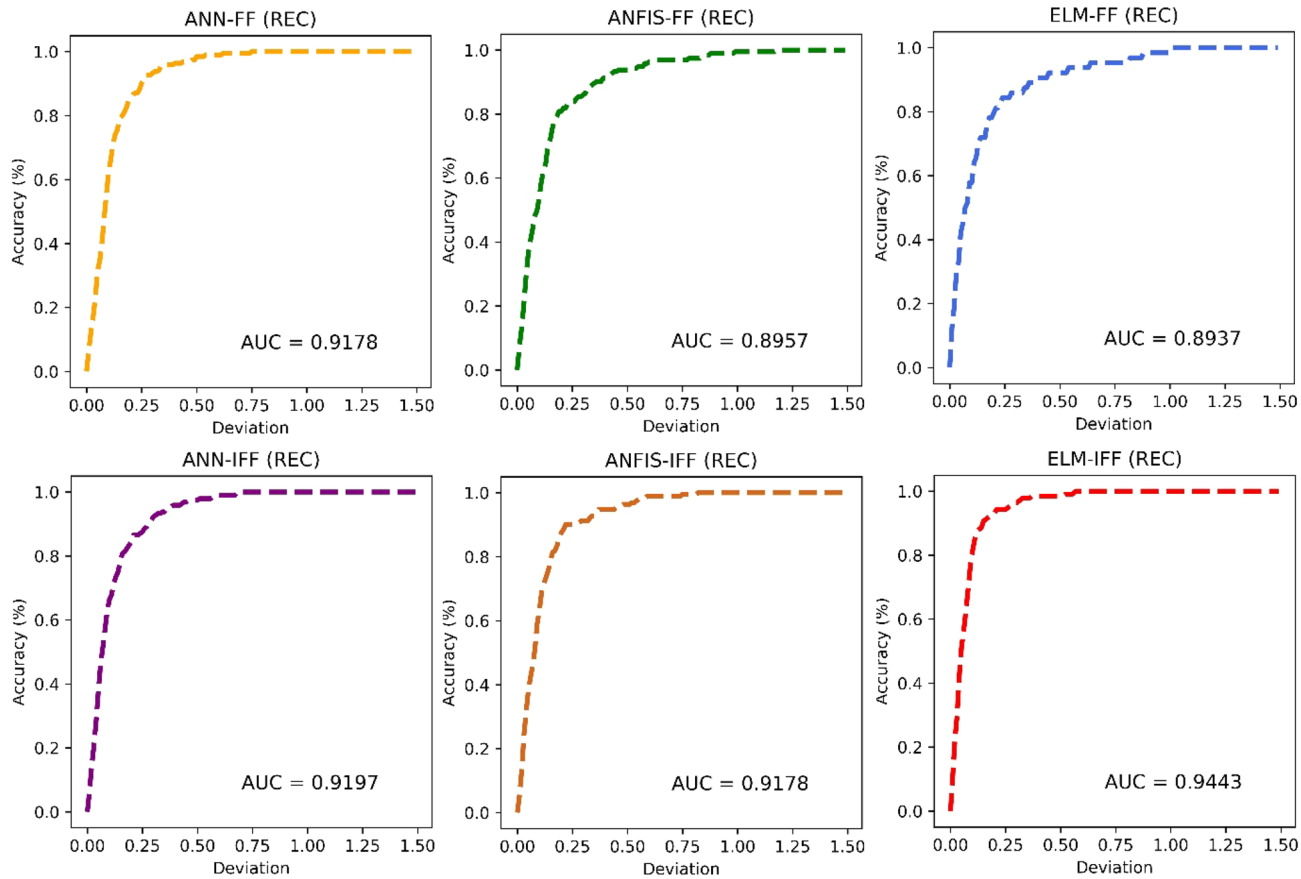


Fig. 10 REC plots of hybrid models for the training stage

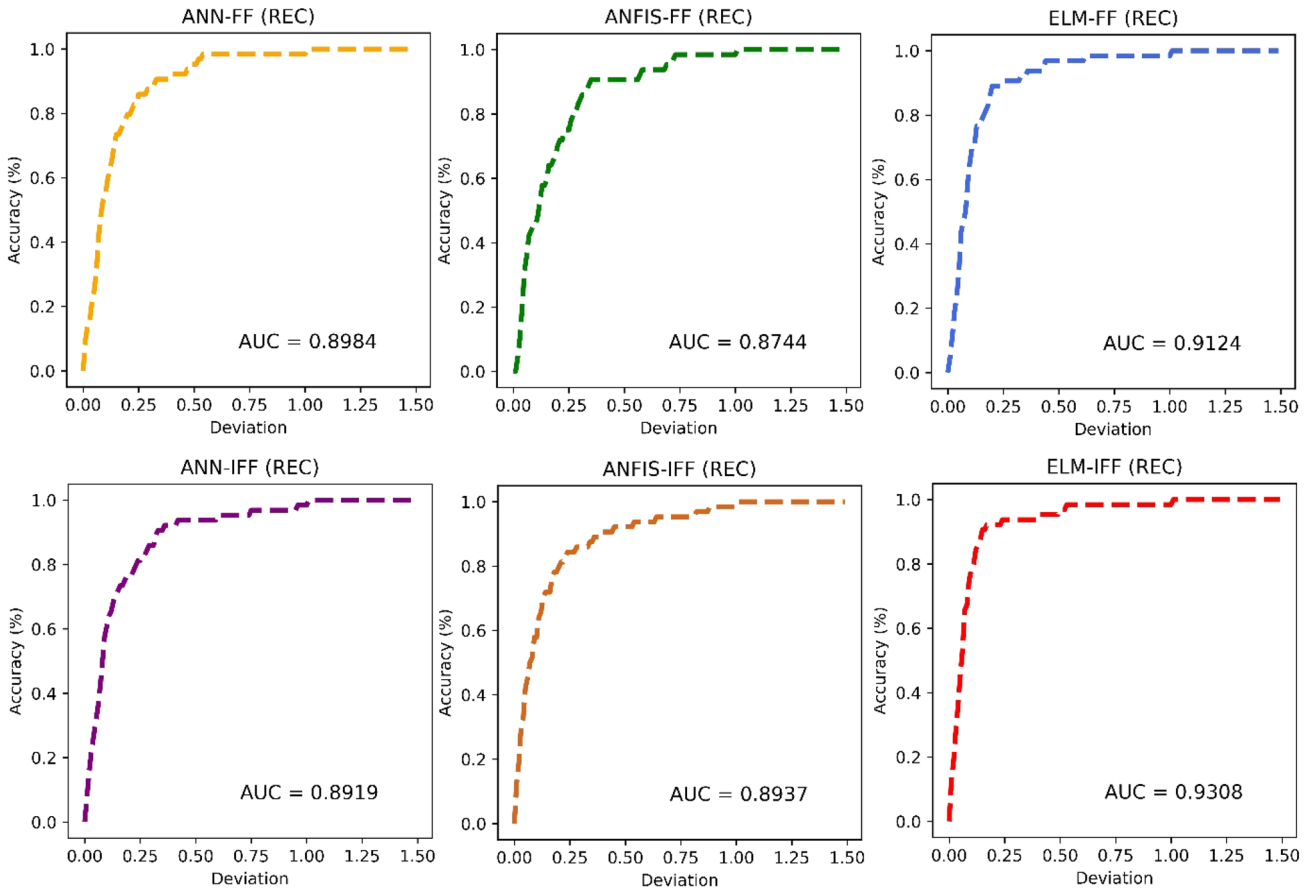


Fig. 11 REC plots of hybrid models for the testing stage

array X defined as, $X = \{X_1, X_2, X_3, \ldots X_i, \ldots, X_n\}$. Each of the variable X_i, in the data array X is a vector of length of m i.e., $X_i = \{X_{i1}, X_{i2}, X_{i3}, \ldots X_{im}\}_1$. The relationship between the strength of the relation (R_{ij}) and datasets of x_i and x_j can be calculated using the following equation:

$$R_{ij} = \frac{\sum_{k=1}^{m} (x_{ik} \times x_{jk})}{\sqrt{\sum_{k=1}^{m} (x_{ik})^2 \times \sum_{k=1}^{m} (x_{jk})^2}}. \quad (20)$$

The results of SA for the estimation of k of soils are presented in Table 8 and Fig. 13. The information presented in Table 8 and Fig. 13 illustrates that the most influential input variables in estimating the k of soils is the QC, followed by γ_d, SC, S, n, and CC for ELM-IFF model. The impact of the input variables on k of soils for the other developed models can be obtained from Table 8.

This study further investigates the widely used Akaike information criterion (AIC) [58–60] to compare the performance of the developed models regarding accuracy and complexity simultaneously. The expression for calculating the AIC is given by:

$$\mathrm{AIC} = 2k - 2\ln\left(\sqrt{\frac{1}{n}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2}\right), \quad (21)$$

where k is the number of adjustable parameters, y_i and $\hat{y}_i$ are the actual and predicted values, N represents the number of samples. The results of the AIC analysis are presented in Table 9. The superior model will have the smaller lower value of this index. As can be seen, the proposed ELM-FF and ELM-IFF attained lower AIC value indicate superior model for predicting the k of soils.

4.6 Comparison with empirical models

In this section, the obtained results from the proposed hybrid models in this study are compared to three empirical prediction models for thermal conductivity of soils. The concept of normalised thermal conductivity, i.e. k_r, is introduced by Johansen [61] as below:

$$k = k_r(k_{sat} - k_{dry}) + k_{dry}, \quad (22)$$

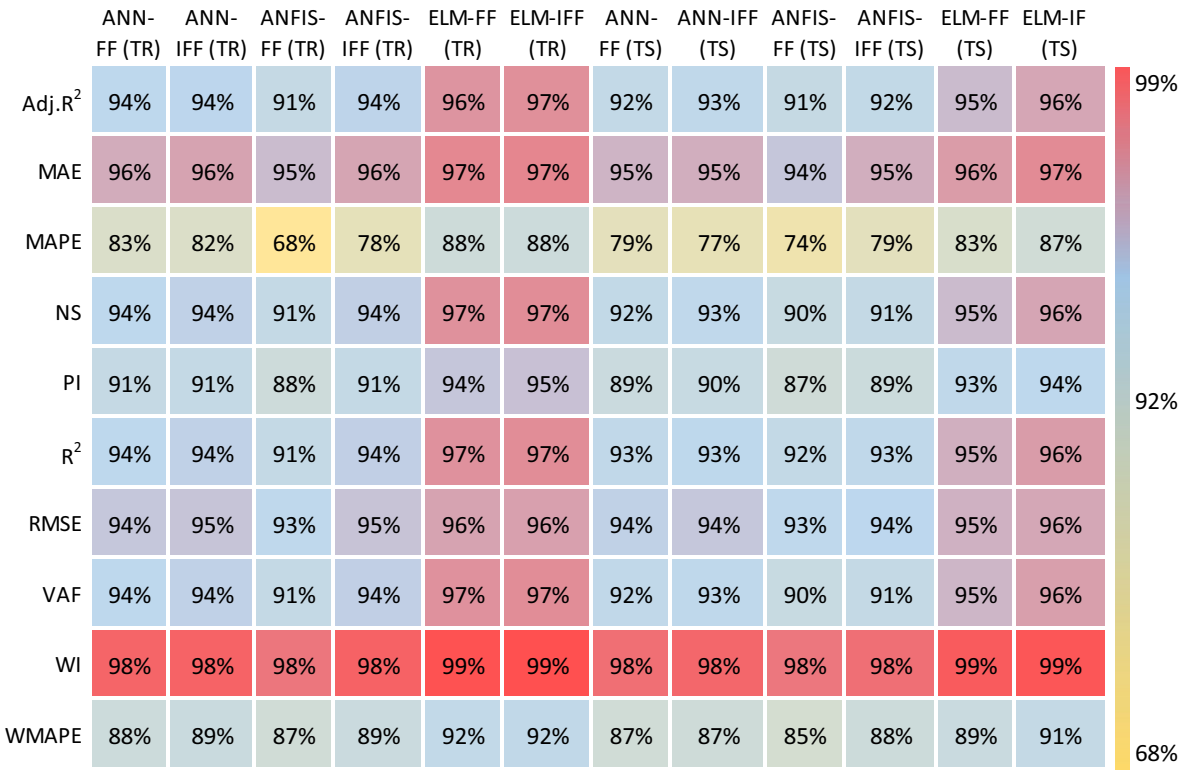


Fig. 12 Accuracy Matrix of the hybrid models for training and testing stages

Table 6 Rank analysis of the hybrid models for the training stage

Performance parameters	Training set					
	ANN-FF	ANN-IFF	ANFIS-FF	ANFIS-IFF	ELM-FF	ELM-IFF
AdjR^2	2	3	1	4	5	6
MAE	2	4	1	3	5	6
MAPE	4	3	1	2	5	6
NS	2	3	1	4	5	6
PI	2	3	1	4	5	6
R^2	2	3	1	4	5	6
RMSE	2	3	1	4	5	6
VAF	2	3	1	4	5	6
WI	2	3	1	4	5	6
WMAPE	2	4	1	3	5	6
Total rank	22	32	10	36	50	60

where thermal conductivity of soils in dry and fully saturated conditions are shown by k_{dry} and k_{sat} ($wm^{-1}\ k^{-1}$), respectively. For totally dry soils the saturation degree (SD) is zero, hence k is equal to k_{dry} and k_r is equal to zero. Whereas, for fully saturated soil SD is equal to 1, so k is equal to k_{sat} and k_r is equal to 1. Three empirical models which use this concept for predicting the thermal conductivity of soils are as below:

Cote and Konard [7]:

$$k_r = \frac{\kappa S}{1+(\kappa-1)S}\begin{cases}\kappa = 4.6 & \text{gravel/coarse sand}\\ \kappa = 3.55 & \text{fine sands}\\ \kappa = 1.9 & \text{silt and clay}\end{cases}. \quad (23)$$

Table 7 Rank analysis of the hybrid models for the testing stage

Performance parameters	Testing set					
	ANN-FF	ANN-IFF	ANFIS-FF	ANFIS-IFF	ELM-FF	ELM-IFF
AdjR^2	2	4	1	3	5	6
MAE	2	4	1	3	5	6
MAPE	3	2	1	4	5	6
NS	3	4	1	2	5	6
PI	3	4	1	2	5	6
R^2	2	4	1	3	5	6
RMSE	3	4	1	2	5	6
VAF	3	4	1	2	5	6
WI	3	4	1	2	5	6
WMAPE	2	3	1	4	5	6
Total rank	26	37	10	27	50	60

Table 8 Results of Sensitivity analysis for the developed models

Models	γ_d	n	S	QC	SC	CC
ANN-FF	0.9103	0.5949	0.8177	0.9290	0.8944	0.3299
ANN-IFF	0.9080	0.5913	0.8194	0.9282	0.8975	0.3180
ANFIS-FF	0.9052	0.5772	0.8108	0.9341	0.8888	0.3201
ANFIS-IFF	0.9036	0.5867	0.8203	0.9287	0.8911	0.3133
ELM-FF	0.9063	0.5913	0.8163	0.9261	0.8932	0.3221
ELM-IFF	0.9067	0.5931	0.8165	0.9270	0.8939	0.3228

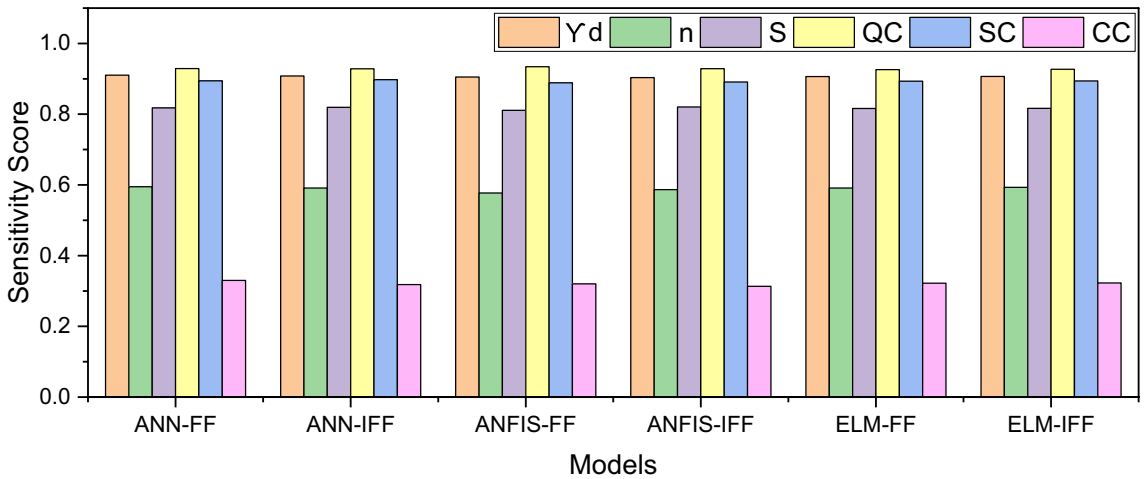


Fig. 13 Sensitivity analysis of six input parameters on thermal conductivity of the soil

Barry-Macaulay et al. [9]:

$$k_r = \frac{\kappa S}{1+(\kappa-1)S}\begin{cases}\kappa = 6.5 & \text{coarse sand}\\ \kappa = 5.6 & \text{fine sands}\\ \kappa = 2.2 & \text{silt and clay}\end{cases}. \tag{24}$$

Lu et al. [8]:

$$k_r = \exp\left[\alpha\left(1-S^{\alpha-1.3}\right)\right]\begin{cases}\alpha = 0.96 & \text{coarse soils}\\ \alpha = 0.27 & \text{fine soils}\end{cases}. \tag{25}$$

Table 9 Details of AIC value

Particulars	ANN-FF	ANN-IFF	ANFIS-FF	ANFIS-IFF	ELM-FF	ELM-IFF
AIC	167.7821	167.8315	275.4276	275.8447	146.3865	146.4401

Table 10 Performance comparison of the thermal conductivity models

Predictive models	R^2		RMSE		VAF	
	TR	TS	TR	TS	TR	TS
ANN-FF	0.9375	0.9275	0.0555	0.0593	93.7458	92.3253
ANN-IFF	0.9405	0.9344	0.0542	0.0554	94.0461	93.0874
ANFIS-FF	0.9122	0.9159	0.0663	0.0663	91.1866	89.9928
ANFIS-IFF	0.9413	0.9269	0.0538	0.0618	94.1251	91.3118
ELM-FF	0.9658	0.9541	0.0410	0.0469	96.5817	95.1317
ELM-IFF	0.9676	0.9615	0.0400	0.0428	96.7603	96.0600
Cote and Konard [7]	0.8135	0.7945	0.0943	0.0932	82.3487	80.8567
Barry-Macaulay et al. [9]	0.8317	0.8261	0.0821	0.0832	83.3462	83.0231
Lu et al. [8]	0.7623	0.7784	0.1023	0.1077	77.3265	77.5687

Table 10 depicts a summary of the models' performances in a comparative manner. As can be seen, proposed models show the absolute superiority, ELM-IFF particularly, over the empirical models.

4.7 Discussion on results

The presented metaheuristic models are evaluated in every possible way to estimate the thermal conductivity of soils. The assessment is initially conducted by employing ten performance indices, including R^2, adj.R^2, VAF, PI, WI, NS, RMSE, MAE, MAPE, and WMAPE. The ELM-IFF model obtained the highest predictive performance with ($R^2 = 0.9676$ and RMSE = 0.04) and ($R^2 = 0.9615$ and RMSE = 0.0428) in the training and testing stages, respectively. The performance parameter details for the models are reported in Tables 5 and 6. It is concluded that there have been found no meaningful uncertainties among the outcomes; neither significant deviations, nor undesirable values. Additionally, visualisations of outcomes in the form of the Taylor diagram, REC curve, and accuracy matrix were provided and discussed. In terms of R^2, RMSE, and the ratio of variances, the Taylor diagrams provided a succinct statistical description of the models. In addition, the REC curve and accuracy matrix display the cumulative distribution of the squared error and the degree of inaccuracy, respectively. Finally, the best predictive model performance was evaluated based on a novel approach known as the "Rank Analysis". The ELM-IFF model attained the highest final rank of 120 while the ANN-FF received the lowest value. This is primarily related to the ability of the IFF algorithm to maintain a balanced approach, both in the phases of exploration and exploitation. It explores the whole space of potential solutions before converging to the global optimum. The ELM-FF model reached the final rank of 100, which is reasonable. It may also be concluded that the IFF algorithm is indeed helpful in the prediction of soil thermal conductivity by optimising the ANN, ANFIS, and ELM models.

5 Conclusion

To predict the thermal conductivity of soils, six novel metaheuristic hybrid models, including ANN-FF, ANN-IFF, ANFIS-FF, ANFIS-IFF, ELM-FF, and ELM-IFF were developed and validated in this study. Experimental test data was gathered from the literature for soils of various classes. The developed models including the combinations of two metaheuristic optimisation approaches (FF and IFF) and ANN, ANFIS and ELM were applied to establish a mapping function to infer the thermal conductivity using a collection of influential factors. In addition, two optimisation techniques, FF and IFF were used to optimise weights and biases of ANN, ANFIS and ELM. First, the main dataset was segmented into training and testing sets. Then, the training data were employed to create the optimisation models based on ANN, ANFIS and ELM. The testing set, however, was applied for validation objectives. Performance indices findings confirm that in the testing phase, the ELM-IFF model obtained the highest predictive efficiency with $R^2 = 0.9615$, VAF = 0.9606, RMSE = 0.0428 and MAE = 0.0316. The "Rank Analysis" also revealed that, at all points, the ELM-IFF model outperforms the others. This result is substantially better than other models based on ANN, ANFIS, and ELM-based models, including ANN-FF, ANN-IFF, ANFIS-FF, ANFIS-IFF, and ELM-FF. The ELM-IFF model is a promising tool in predicting thermal conductivity of soils based on the outcomes and the evidence.

This study is aimed not only to replace the actual test operation to determine the thermal conductivity of soils but to provide details about the prior AI-based studies indicating the findings at all levels. Six metaheuristic soft computing-based techniques were therefore constructed to generate robust and efficient predictive models. The key feature of the developed models is their high generalisation potentials, negligible over-fitting concerns, and very low computational costs. The performance of the models was found to be in good agreement in the testing phase compared to the performance in the training phase. During the testing phase, neither considerable variations nor undesirable values were obtained, which fulfils the generalisation capability and robustness of the model. Given the overall performance, the ELM-IFF model can be introduced as a promising alternative to assist geotechnical professionals in approximating the thermal conductivity of soils. Furthermore, and to authors' best knowledge, this study is the first application of ANN, ANFIS, and ELM-based techniques hybridised with an improved firefly optimisation algorithm (IFF) in predicting the thermal conductivity of soils.

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